Automatic Parameter Learning for Easy Instruction of Industrial Collaborative Robots

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***Abstract*—The manufacturing industry faces challenges in meeting requirements of flexibility, product variability and small batch sizes. Automation of high mix, low volume productions requires faster (re)configuration of manufacturing equipment. These demands are to some extend accommodated by collaborative robots. Certain actions can still be hard or impossible to manually adjust due to inherent process uncertainties.**

This paper proposes a generic iteratively learning approach based on *Bayesian Optimisation* to efficiently search for the optimal set of process parameters. The approach takes into account the process uncertainties by iteratively making a statistical founded choice on the next parameter-set to examine only based on the prior *binomial outcomes*. Moreover, our function estimator uses *Wilson Score* to make proper estimates on the success probability and the associated uncertain measure of sparsely sampled regions. The function estimator also generalises the experiment outcomes to the neighbour region through kernel smoothing by integrating *Kernel Density Estimation*.

Our approach is applied to a real industrial task with significant process uncertainties, where sufficiently robust process parameters cannot intuitively be chosen. Using our approach, a collaborative robot automatically finds a reliable solution.

***Index Terms*—industrial assembly, parameter optimisation**

I. Introduction

For the past decades, industrial robots have been used for the automation of high volume industrial assembly tasks. In high volume production, e.g. the automotive industry, efforts are made to optimise robot motions, to ensure a structured environment, to eliminate process uncertainties, and to reduce the cycle time. Recently, the automation of low volume tasks has caught increasing interest. To make such assembly tasks cost-effective, a faster and easier configuration of the robot set-up is required. *Collaborative robots* can facilitate this through their more intuitive interfaces and improved safety features.

In terms of robot programming, one approach is to create a higher-level, task-focused abstraction for online robot programming in order to both expedite the programming and reduce the need for robotics expertise. One such approach is the skill-based programming proposed by [1]; specifically intended for collaborative robots. Here, expert knowledge is encapsulated into parametric "skills" which represent task-related capabilities of the robot system. Hence, programming a task is done by sequencing these skills followed by an online parameter instantiation through kinesthetic teaching.

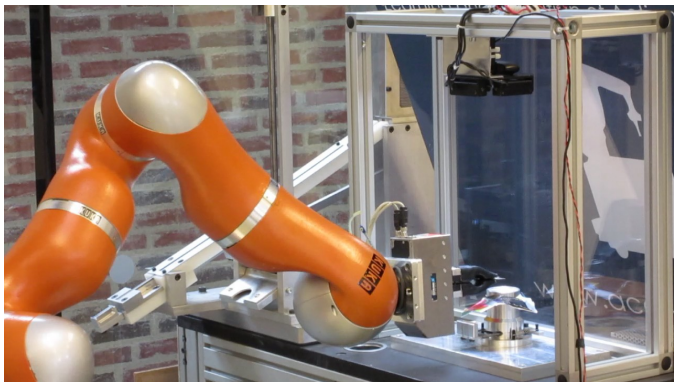


Fig. 1: The laboratory set-up used for the experiments in this paper. The robot gradually learns to drop a metal ring into the fixture so the ring settles correctly. The camera above the fixture is used to automatically evaluate each attempt.

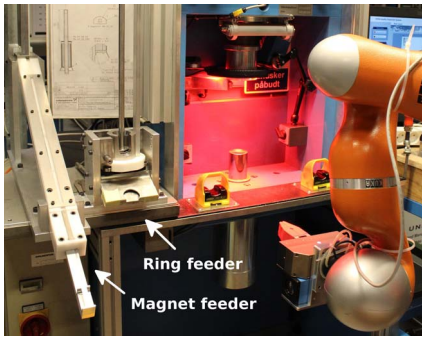
In low volume productions, it is necessary to use standard fixtures and tools, as the design and construction of specialised, dedicated equipment is often too time-consuming and expensive. However, this potentially introduces more uncertainties into the process which for certain actions in the assembly task can influence the execution. If not taken into account, these uncertainties can result in a lower robustness which is undesirable since it demands the operator to more frequently interfere. However, handling this process uncertainties can even for expert users be hard or even impossible when instructing the task through online programming.

A solution to overcome process uncertainties is to use active compliance such as force-torque sensing from which the robot motion can actively be adapted during execution. However, these solutions raise the cost and tend to increase the cycle time. Instead, we aim for a solution where the robot runs blindly based on a predefined motion for which the difficult part of the assembly has been learned in advance.

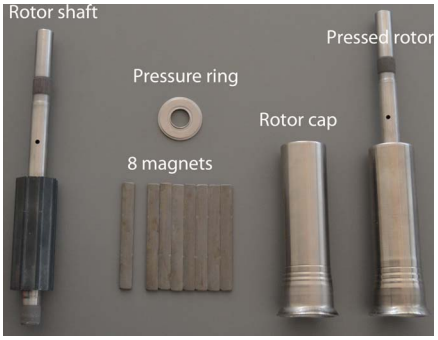
In this paper, we present a learning-based approach for managing the process uncertainties in assembly cases. The approach is centred around a machine learning solution that for a given assembly action seeks the set of parameters which

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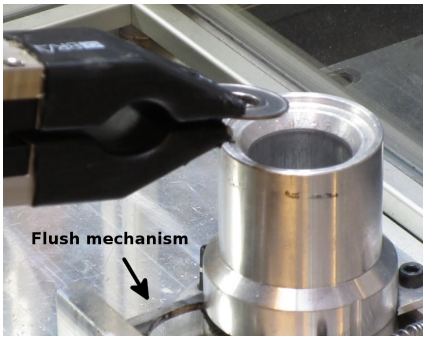
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(a) Press station at Grundfos A/S.



(b) Parts to be assembled.



(c) Press fixture mock-up.

Fig. 2: The case treated in this paper is from the leading manufacturer of circulation pumps, Grundfos A/S. The press station in (a) is used to assemble and combine the 11 parts shown in (b). The case is described in more details in [2] where also (a) and (b) origins. In (c), a close up of the press fixture which was used for the experiments in this paper is shown.

gives the highest chance of success. The purpose of this work is to amend online programming methods for collaborative robots with the ability to manage process uncertainties through automatic learning and thereby increase the success rate. Hence, the operator still instructs the task using e.g. a skill-based programming approach and kinesthetic teaching, but afterwards the robot conducts a learning cycle on the assembly actions challenged by the process uncertainties.

A. Industrial Test Case

A real industrial case is used as a test bed for the proposed learning approach (see Fig. 2). The case chosen is an assembly task from the pump manufacturer Grundfos A/S. The task consists of manually assembling four parts inside a press, activating the press and removing the final product. This assembly has not been automated because the production volume is too small to justify the investment of a fully automated production line. This is similar to many other cases found in the industry. In the European FP7 project TAPAS [3] it was shown that collaborative mobile robots *can* be employed to solve this and related tasks, although certain parts which require a high degree of precision were difficult and inconveniently time-consuming to set up. For a more thorough case description, please refer to [2].

The work-cell consists of a KUKA LWR4+ robot arm on which a Schunk WSG50 parallel gripper is attached (see e.g. Fig. 1). The work-cell also contains part feeders for both the rings and magnets (see Fig. 2a) and a fixture in which the assembly takes place (see Fig. 2c). The robot can mechanically actuate both feeders and flush the mock-up of the fixture using the actuating mechanism on the front.

The assembly procedure is as follows:

1) Drop a ring into the fixture such that it settles in the correct final position (as shown in Fig. 2c).
2) Place a rotor shaft into the fixture and through the ring.
3) Insert 8 magnets into the slits in the rotor shaft.
4) Place the rotor cap over the rotor shaft.

Most of these operations proved relatively easy to program on a collaborative robot using skill-based programming and manual kinesthetic movement of the robot arm [2]. However, dropping the ring proved challenging, since very small differences in the way the ring is dropped can cause the success rate to change drastically. Hence, the optimal solution is not intuitively chosen. The process uncertainties which lead to the small differences are caused both by the feeding equipment and by the gripper fingers.

It would significantly simplify the programming of the whole task if the parameters in the ring operation could be learned automatically. We believe that this generalises to several tasks found in the industry [4]. Therefore this case is chosen as a demonstration case for the "machine learning on collaborative robots"-approach proposed in this paper.

II. State of the Art

One recent project dealing with improving the general capabilities of collaborative manufacturing and logistic robots through learning is *RoboEarth* [5]. The authors in [5] propose to learn data on individual robots and share the learned knowledge automatically over the cloud. A skill abstraction layer makes the learning generic and thus usable on different platforms. However, while learning is part of the architecture, focus in their project is on the knowledge sharing and generalisation across different platforms.

The authors have previously proposed a concept for generic robot skills supporting both autonomous and manual planning and instantiation [6]. The work focuses on usability, and several experiments have shown its potential in industrial scenarios [2], [7]. Learning has, however, not previously been explored in this context, even though the framework and method can easily be extended to support this.

Other authors have investigated how learning can be used to generalise movements based on force/torque profiles [8], [9], however, the approaches are limited to deal with forces and torques along trajectories. Contrary to this, the approach proposed in this paper uses a more generic method, which also can include sensing and vision in its learning pipeline.

The work in [10] investigates the insertion of a battery into a cell phone where process uncertainties can lead to an

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unsuccessful assembly. To comply with these uncertainties, the authors optimise the process parameters using an iterative approach. The approach always chooses the set of parameters with the highest upper limit of the confidence interval which is calculated by Wilson Score. This is partly similar to the approach used in this paper, however, we extend the calculation of the confidence interval by using Kernel Density Estimation to generalise the experiment outcomes to the neighbour region of the parameter space.

III. Method

The aim for our learning method is to find the set of parameters with the highest success probability. Therefore, the process has to be represented by some parameters which can be optimised according to a pre-defined success criterion. This section briefly describes the learning approach, whereas, section V defines the concrete parameters and success criterion.

Our learning approach is based on Bayesian Optimisation [11] and does in each iteration consist of three steps:

1) **Select** the next parameter-set, x_i, to investigate.
2) **Perform** an experiment with the parameter-set x_i and obtain an outcome, o_i.
3) **Update** the success probability estimates and the associated uncertainty measures based on o_i.

In addition, an experiment can either be evaluated a success or a failure such that $o = \{1, 0\} = \{s, f\}$. This choice is made since it is often much harder to judge how well the assembly succeeded (e.g. $[0, 1]$). In the following, we will explain how the next sample parameter-set is chosen, and how the function estimator which gives the success probability estimate and associated uncertainty measure works.

A. The Function Estimator

To update the success probability estimate and the associated uncertainty measure we use Wilson Score (WS) [12] extended with Kernel Density Estimation (KDE) [13] as function estimator abbreviated WS-KDE. WS gives a proper estimate even in situations where only a few samples have been performed. However, WS is a discrete estimator and does therefore only give an estimate for a specific parameter-set. To compensate this we use KDE which by its kernel smoothing generalized the outcome of an experiment to the neighbour region. We have in [14] shown that by using KDE as function estimator (not combined with WS), the number of samples needed for finding a solution can be reduced.

1) The discrete Wilson Score estimator: The WS discrete estimator of the success probability is defined as in [12]:

$$\hat{\mu}_{ws} = \frac{\hat{\mu}_{na} + \frac{1}{2n}z^2}{1 + \frac{1}{n}z^2}, \tag{1}$$

where $\hat{\mu}_{na} = \frac{1}{n}\sum_{i=1}^{n} o_i$, n is the number of samples made in the specific parameter-set, and z is the $(1 - \frac{\alpha}{2})$ quantile for a two-sided interval.

The WS discrete uncertainty measure is defined as in [12]:

$$um_{ws} = \frac{z}{1 + \frac{1}{n}z^2}\sqrt{\frac{1}{n}\hat{\mu}_{na}(1 - \hat{\mu}_{na}) + \frac{1}{4n^2}z^2}. \tag{2}$$

2) The KDE estimator: To include KDE in WS, $\hat{\mu}_{na}$ in (1) and (2) is replaced with the KDE success probability estimate which is defined as:

$$\hat{\mu}_{kde}(x) = \frac{n_{glo}^{-1}\sum_{i=1}^{n_{glo}} K_{h,x_i}(x)\, o_i}{n_{glo}^{-1}\sum_{j=1}^{n_{glo}} K_{h,x_j}(x)}, \tag{3}$$

where K_{h,x_i} is the kernel with bandwidth, h, and n_{glo} is the total number of samples in the whole parameter space, $\mathcal{X}$.

3) The number of samples: We replace n in (1) and (2) by:

$$n(x) = \frac{n_{glo}h}{||K||_2^2}\hat{f}_h(x), \tag{4}$$

where $||K||_2^2$ is the squared L_2 norm of an identity kernel, and $\hat{f}_h(x) = \frac{1}{n_{glo}}\sum_{i=1}^{n_{glo}} K_{h,x_i}(x)$ is the estimated density.

Hence, we obtain an estimate on the success probability with an associated uncertainty measure based on KDE and WS which in the parameter space generalises the outcome of an experiment to the neighbourhood and corrects the estimates of sparsely sampled regions respectively. In addition, the estimated success probability and uncertainty measure can be combined to the confidence interval as $[\,\mu(x) \pm um(x)\,]$.

B. The Selection Method

The first step in each iteration of our learning approach is to select the next parameter-set to investigate. This is done by the Upper Confidence Bound (UCB) (see e.g. [15]), which selects the parameter-set with the highest upper bound of the confidence interval. In this way, the parameter-set with the current highest potential is chosen. In each iteration, the parameter-set to investigate is selected as follows:

$$x_i = \underset{x \in \mathcal{X}}{\operatorname{argmax}}\left(\mu(x) + um(x)\right), \tag{5}$$

where x is a parameter-set in the parameter space, $\mathcal{X}$. Moreover, $\mu(x)$ is the estimated success probability and $um(x)$ is the associated uncertainty measure. Using (5) indirectly makes a trade-off between exploration and exploitation. The uncertainty measure, $um(x)$, overall gradually decreases by the increased number of samples performed in the neighbour region. Note, when using Gaussian Processes [16], UCB is often defined as $\mu(x) + \kappa\sqrt{\sigma^2(x)}$, where $\mu(x)$, $\sigma^2(x)$ and κ is the mean, variance and exploration-exploitation trade-off.

IV. Experimental and Method Set-up

We will in this paper examine how to drop the ring so it settles facing down in the fixture as shown in Fig. 3. However, the ring can be dropped either facing down or up depending on its orientation in the feeder. We apply our learning algorithm to both cases and examine the results and differences. In addition, the figure also describes the three parameters (height, distance and angle) that defines the drop position of the ring, and how the ring finally should be orientated when settled in the fixture. The bounds of the parameters are for the height, distance and angle chosen to $[2; 30]\,mm$, $[14; 30]\,mm$ and $[0; 30]^\circ$ Note, the *height* must at least be $2\,mm$ due to the thickness of the ring, and the *distance* cannot be lower than $14\,mm$ since the fingers otherwise will collide with the fixture.

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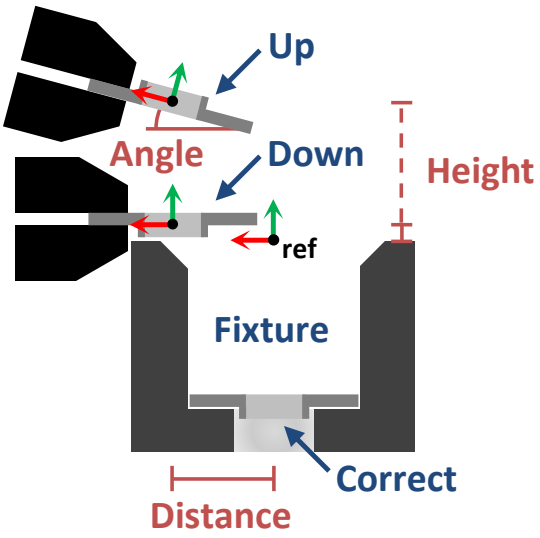


Fig. 3: Schematic of the ring case showing the two drop positions (up and down), the correct final position of the ring, and the three parameters (height, distance and angle) given relative to the reference coordinate system.

For the iterative learning approach, each experiment needs to be labelled as either a success or a failure. This is done by a simple and inexpensive vision set-up using a standard web-camera (see Fig. 1), which from the top can inspect the fixture (see Fig. 4). By common template matching, it can be concluded if the ring is correctly orientated in the fixture.

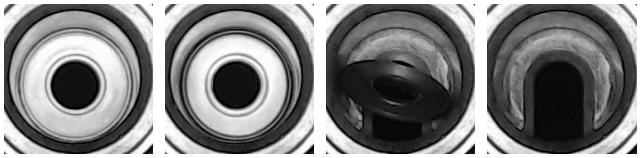
The selection of the next parameter-set in each iteration is always done using (5), but to ease the selection a discretised parameter space is used. Hence, an assent algorithm is not required to maximize (5), however, the achievable resolution is fixed by the grid. The discretisation is chosen to $\Delta = [\,1\,mm\,,\,1\,mm\,,\,5^{\circ}\,]$ for the height, distance and angle respectively. The kernel bandwidth, h, becomes a matrix for multi-dimensional problems, which empirically is chosen to $H = diag[\,2\,mm\,,\,2\,mm\,,\,5^{\circ}\,]$. Moreover, 500 iterations are conducted for both the ring facing down and up, and we define the *best parameter-set* as the parameter-set with highest success probability obtained hereafter. The drop position is reached using the robots own Cartesian position control.

V. Experimental Results and Discussion

This section presents and discusses the results obtained from the experiments individually conducted for both the ring facing down and up. First, the best parameter-set obtained for each case is examined in relation to the effect of the process uncertainties. Secondly, the upper confidence bound and the estimated success probability through the iterations are inspected and linked together with the function estimator.

A. The Best Parameter-Set for Each Ring Case

The results from applying our learning approach to both ring cases are shown in Table I. The table contains for each case, the best parameter-set, x_{best}, found along with its success probability estimates by WS-KDE and the number of trials in that specific parameter-set. Please note, that also experiments carried out in the neighbourhood affects the probability estimate in this point due to the kernel smoothing. Moreover, Fig. 5 shows for each ring case a cross-sectional view of the parameter space where the best parameter-set is marked red.



(a) Facing down. (b) Facing up. (c) Skewed. (d) No ring.

Fig. 4: The potential outcomes when dropping the ring into the fixture seen from the vision system (see Fig. 1). In (a) the ring is correctly facing down, however, in (b) it is wrongly facing up which can be seen by the slightly darker outline of the ring in the fixture. The ring can also remain standing as shown in (c). The empty fixture is shown in (d).

Table I shows that for the ring facing down (same as the final assembly orientation of the ring), the best parameter-set, x_{best}, obtains a success probability of $94.2 \pm 1.5\,\%$. Moreover, this parameter-set is located on the boundary of the chosen parameter space (see Fig. 5a). As a result of having the lowest possible height, the ring is released on top of the fixture and consequently slides into the fixture. Sliding the ring into the fixture instead of dropping it will minimise the effect of the stochasticity in the process, since the random displacement is likely to be increased when the ring bounces on the fixture.

Examining the results for the ring facing up shows that the ring should be dropped from $15\,mm$ with an angle of 15° (see Fig. 5b) to obtain the highest success probability at $77.5 \pm 5.8\,\%$. Compared to the ring facing down, the ring now needs to turn around after being dropped to achieve the correct final orientation in the fixture, which decreases the success probability due to the effect of the stochasticity.

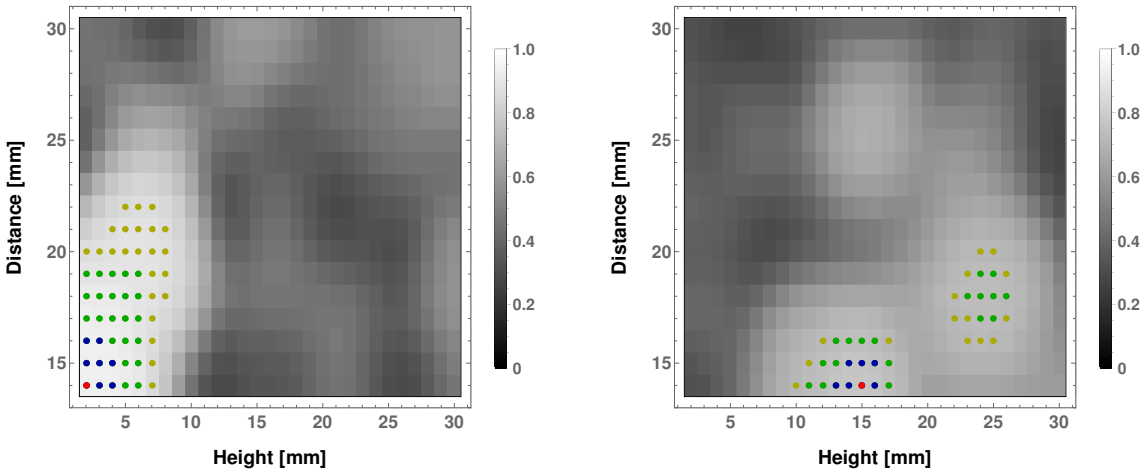
From a manufacturer perspective a success rate at $100\,\%$ would be ideal. Therefore, having a maximum success probability at $94.2\,\%$ would demand that the fixture can be emptied without interaction from the operator, so that the assembly process automatically can continue. This is possible for the test set-up used (see Fig. 2c). Such solution might be acceptable since the ring is the first part of the assembly.

B. The Upper Confidence Bound for each Ring Case

For the ring facing down and up, Fig. 6 shows the upper confidence bound and the estimated success probability (and hence also the uncertainty measure) for the parameter-set chosen in each iteration. In addition, the two figures also show when the final best parameter-set, x_{best}, is sampled.

For the case where the ring is facing down (see Fig. 6a), the learning approach explores the parameter space in the beginning, but hereafter quickly finds and mostly focuses on the parameter-set which in the end is found to be the best,

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(a) Ring facing down for an *angle* of 0°.

(b) Ring facing up for an *angle* of 15°.

Fig. 5: Cross-sectional views of the estimated success probability for the parameters *height* and *distance* at a certain angle (see also Table I and Fig. 3). The red point in each plot represents the best parameter-set found by the learning approach after the 500 iterations. The blue, green and yellow points represent the 10, 50 and 100 best parameter-sets for the entire parameter space and, therefore, are all these points not shown in the above two cross-sectional views.

TABLE I: The obtained results found by the iterative learning approach when applied to the two ring cases.

Face	**The best parameter-set (x_{best})**			**The estimates by WS-KDE for x_{best}**			**Number of trials for x_{best}**		
	Height	Distance	Angle	Success Probability	Uncertainty	Confidence Interval	Total	Successes	Failures
Down	$2\,mm$	$14\,mm$	0°	94.2 %	±1.5 %	[92.7; 95.7] %	285	270	15
Up	$15\,mm$	$14\,mm$	15°	77.5 %	±5.8 %	[71.7; 83.3] %	42	37	5

x_{best}. Many successes are quickly obtained in this parameter-set due to the relatively high success probability. This entails that the UCB value in this point is slightly higher than the UCB value for regions which are even only peripherally influenced by failures. Hence, the parameter-set is continuously chosen until the influence of failures drags its UCB value below the others. That the best parameter-set is heavily sampled is shown in Fig. 6a, and consequently leads to a rather low uncertainty of the estimated success probability (see Table I).

For the case where the ring is facing up, the success probability is lower than for the ring facing down (see Table I). As a consequence of the lower success probability many other parameter-sets are potentially better and therefore tried, since they are unexplored and thereby have a higher UCB value. The reason for this is that Wilson Score adjusts the confidence interval towards $[0.5 \pm 0.5]$ (see Section III-A) when few experiments have been made (a low $n(x)$). Because of the lower success probability, other regions need a higher negative influence (more failures in the neighbour region) to be proven worse than the final best parameter-set. That other parameter-sets are tried can be seen from the many spikes in Fig. 6b compared to Fig. 6a. Because of the higher exploration, fewer trials have been performed in the final best parameter-set (see Table I). Moreover, Fig. 6b also shows that conducting more iterations for the ring facing up could be beneficial to get a more reliable estimate of success probability since the final best parameter-set has an uncertainty measure of $\pm 5.8\,\%$.

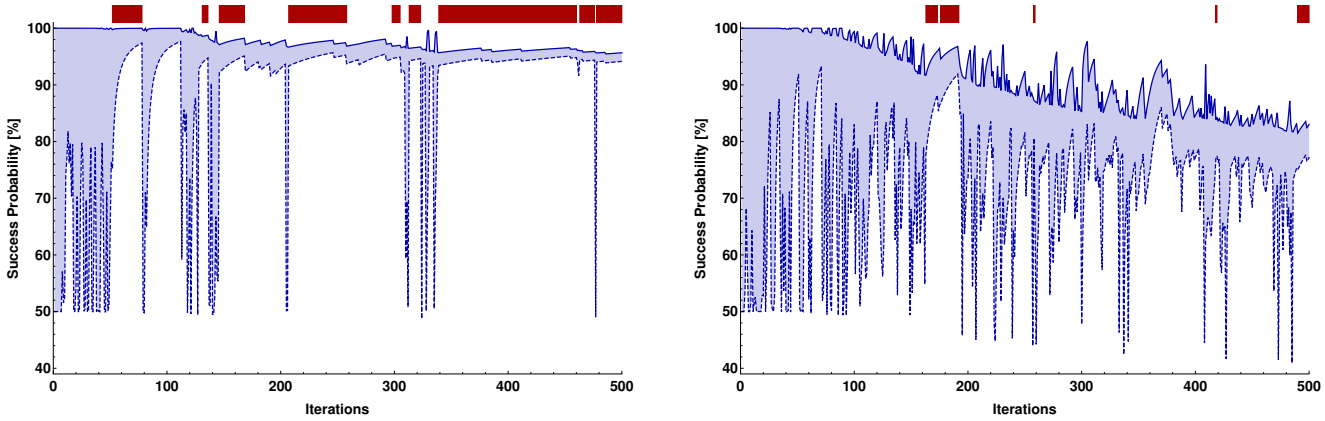
C. The Number of Experiments and the Uncertainty Measure

From Fig. 6 it can be seen that few experiments are initially needed to quickly reduce the uncertainty measure of the estimated success probability (see e.g. Fig. 6a from iteration 52 to 78), however, rather many experiments are additionally needed to obtain a very low uncertainty measure (see e.g. iteration 339 to 460). As an example, Wilson Score without the KDE extension (see Section III-A) needs 60 experiments for a success probability at $p = 0.8$ to obtain an uncertainty measure less than $\pm 10\,\%$, whereas, 6145 experiments are needed for an uncertainty measure less than $\pm 1\,\%$. Hence, it might be beneficial to define the lower bound of the confidence interval as a threshold for accepting (and thereby as a stopping criterion for the learning), than using the uncertainty measure itself.

VI. Conclusion

This paper suggests a generic iteratively learning approach used to find a good parameter-set for assembly operations, which might not be intuitive to manually choose for the robot programmer during online programming. The approach has been applied to an operation of a real industrial assembly task where the success rate is highly influenced by the process uncertainties. To efficiently sample the parameter space, our learning approach uses statistical calculations and comparisons to iteratively make an efficient selection of the next parameter-set to examine. The confidence interval is estimated both by

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(a) Ring facing down.

(b) Ring facing up.

Fig. 6: The estimated success probability (dashed line) and the upper confidence bound (solid line) for the parameter-set chosen in each iteration. The vertical distance between these two lines (hatched area) represents the uncertainty measure. Moreover, the two figures also show in which iterations the final best parameter-set, x_{best}, is sampled (red rectangles). Note, the two figures do not show the lower confidence bound.

making corrections for sparsely sampled regions and by generalising the outcome of the experiments to the neighbourhood by kernel smoothing. Hence, the inherent process uncertainties are taken into account during the optimisation of the process. We showed that the assembly operation investigated could be optimised to a success probability of $94.2\,\%$ with an uncertainty measure of $\pm 1.5\,\%$ which possibly is sufficient.

Although demonstrated on a single industrial assembly operation, the proposed learning approach is easily generalised to other operations influenced by process uncertainties. Integrating our learning approach into an intuitive online programming procedure for collaborative robots, e.g. skill-based programming using kinesthetic teaching, would ease the instruction of assembly tasks influenced by process uncertainties. Hence, the operator instructs the assembly operation as any other by either defining the allowable parameter space or by kinesthetic teaching. Hereafter, the system automatically conducts the learning. As a result, the instruction time is minimally prolonged, while the robustness of the task is increased.

ACKNOWLEDGMENT

This work was supported by Innovation Fund Denmark as a part of the project "MADE — Platform for Future Production". The authors would like to thank Grundfos A/S for providing the industrial test case used in this work.

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